



**Sultan Qaboos University
College of Engineering
Department of Mechanical and Industrial Engineering**

Final Year Project Handbook
Mechanical & Industrial Engineering Programs

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Department of Mechanical and Industrial Engineering

Sultan Qaboos University
College of Engineering

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Introduction to Final Year Project

The final year project (FYP) is one of the requirements for the degree of Bachelor of Science (BSc) in Mechanical and Industrial Engineering. The FYPs are divided into two phases: FYP1 and FYP2. Each of the projects has to be taken in two separate consecutive semesters.

Mechanical Engineering Program				
Course	Course Code	Credits	Prerequisites	Semester
FYP1	MEIE5193	2	(MEIE4101 or MEIE4102), MEIE4144	Semester 9
FYP2	MEIE5194	3	MEIE5293	Semester 10

Industrial Engineering Program				
Course	Course Code	Credits	Prerequisites	Semester
FYP1	MEIE5291	2	MEIE4285	Semester 9
FYP2	MEIE5292	3	MEIE5291	Semester 10

The FYPs are a powerful tool for integrating and implementing the student's knowledge and skills obtained throughout his study. The topic of the FYPs should cover theoretical and experimental investigation of an engineering problem with a strong emphasis on system design, environmental, economic and ethical issues. The course will focus on detailed design while enhancing students' skills on how to integrate knowledge they gained through their study in the department. Students are expected to tackle open-ended engineering problems whose solutions require design synthesis, judgment, technical skills, analysis, creativity, innovation and cost analysis. More than one faculty member may be involved in the supervision process. In multi-disciplinary projects, team members as well as supervisors from outside the department are encouraged. The main task of the supervisor(s) is to monitor the student work and to properly guide the students throughout the project at regular intervals. At the end of his work, the student is required to submit a report and to deliver a poster presentation (for FYP1) and an oral presentation (for FYP2).

The objectives or rationales for having a final year project for Mechanical and Industrial Engineering students can be summarized as follows:

1. To apply knowledge/skills acquired from previous taken courses.
2. To gain experience in system development.
3. To learn new knowledge and techniques not taught in class.
4. To ensure quality and safety requirements are met

These guidelines are designed to introduce students to the department regulations and policies for final year projects as well as to give guidance on report academic writing for both design-oriented and research-oriented projects.

Detailed FYP Calendar

Final year project students should adhere to the regulations and deadlines regarding the FYP execution. Failure to respect these rules and deadlines will result in grade reduction or possible dismissal from the FYP.

The following important deadlines are designed to follow up the progress of the FYP and ensure its successful completion.

Projects Assignment to Groups		
Tasks	Deadline	Action/Responsibility
Submission of titles and project assignment summary	Week 11	Faculty members
Expression of interest by the students who achieved the pre-requisites to take FYP1	Week 11	Coordinator/Students
Approval of project assignment summary	Week 12	Department Board
FYP Announcement	Week 12	FYP Coordinator
FYP Application Submissions	Week 13	Students
Project Assignment to students	Week14	FYP Coordinator
First Seminar: Final Year Project, An Overview	Week 13/14	FYP1 Coordinators

Projects Implementation		
Tasks	Deadline	Action/Responsibility
Weekly Meeting with Supervisor and Logbook Submission	Weekly	Students
Submission of Project Proposal (for FYP1 only)	Week 5	Students
Submission of Progress Report (for FYP1 and FYP2)	Week 10	Students
Draft Project Report (for FYP1 and FYP2)	Start of Week 14	Students
Final Corrected Project Report (for FYP1 and FYP2)	End of Week14	Students
Draft Poster Design (FYP1)	Start of W14	Students
Final Poster (FYP1)	End of Week 14	Students
Poster/Oral presentations (FYP1/FYP2)	Week 15	Students

FYP Assessment

The Assessment process of FYP Project is a direct responsibility of the supervisor and the examining committee. The assessment would mainly concentrate on the research, analysis and design of the project.

Overall Assessment

The general objective of FYP-I is to have the student performs an academic research on related topics pertaining to the system to be developed and then design the system. However, specific objectives may depend on the project and the agreement between the group and the supervisor.

In general, the evaluation of FYPs will cover the following three aspects:

- **Implementation:** The process/method of conducting the project.
- **Poster/PPT Presentation:** The ability of the group to present their findings and progress using linguistic communication skills as well as technical reasoning.
- **Final Report:** The document summarizing the group's accomplishments.

Based on the three aspects, the evaluation of FYP project will be shared between the project supervisor and the examining committee as appears in the following table:

Criteria	Supervisor	Examiner 1	Examiner 2
Project Implementation	√		
Poster/ oral presentation	√ (draft)		√
Final Report	√ (draft)	√	
Grades share	60%	20%	20%
All the marks are between 0 and 4. The Coordinator may apply the following penalties: • up to 10%: for any delay in submitting the proposal, progress report, etc. • up to 5%: for delay or missing submission of the weekly logbooks • up to 7.5%: for any absentee of students in the weekly meetings			

1. **Project Implementation (60%)**

This portion of the evaluation is aimed at evaluating the way FYP group conducts the project. The following criteria are to be evaluated:

No.	Activity	Weight %
1	The problem statement and project objectives are clearly stated	5
2	Identified realistic constraints (economic, environmental, and ethical) and developed design specifications	10
3	Developed feasible conceptual alternatives and selected the best	15
4	Developed detailed design and drawings using proper analysis (if applicable)	15
5	Adherence to task delivery time schedule	10
6	Attending meeting, participation in team discussion and decision making	10
7	Taking initiative and responsibility for various tasks, Ability to work effectively in a multidisciplinary team	10
8	Maintaining a positive attitude towards team members	5
9	Overall Quality of draft report	15
10	Overall quality of draft Poster/Power Point presentation	10

2. **Final Report (20%)**

The content of the report may vary depending on the type of project undertaken by a group of students. However, since most of the projects in MIE department involve design elements, the marking scheme for the report would be based on nature of project. It is emphasized that report should be written with high quality and should be concise. Length of report should not exceed 100 pages. The suggested marking scheme is shown below.

No.	Activity	Weight %
1	The report writing format follows the FYP guidelines. (This includes quality of figures and tables, captions, equations and citations, references and appendices, if any)	5
2	The language used throughout the report is clear and concise	10
3	The project abstract/summary conclusions, and final findings are comprehensive and clear	5
4	The literature review is relevant up-to-date and well structured	10
5	The references listed are properly cited	10
6	The Project follows proper design methodology:	
	a. The problem statement and objectives are clearly stated	5
	b. Identified realistic design constraints (economic, environmental, and ethical) and developed proper specifications	10
	c. Developed feasible conceptual design alternatives and selected the best design	10
	d. Developed detailed design and drawings using proper analysis (if applicable)	10
7	Relevant professional and design codes are followed (for FYP2 only)	5
8	Decide on the best course of action and justify the decision taking into consideration ethical implications, and after identify possible courses of action. (for FYP2 only)	10
9	Taking into consideration the relevant environmental, economic, health/safety, legal, and ethical impact in the designing process (for FYP2 only)	10

Any report that does not meet the above criteria should be rejected. The student then needs to make the necessary amendments and resubmit the report to the examining committee and the FYP coordinator within two working days from the date of examination. Moreover, any Report that exhibits high level of similarities with other documents (more than 25% with the Turnitin or more than 5% from any single reference) will be also rejected. The Coordinator will take in this, in consultation with the HoD and the Supervisor, the most appropriate decision.

3. **Poster/Final Presentation (20%)**

Each group of students needs to present their work orally to the examiners and any invited attendees (if applicable). The presentation should last for a maximum of 30 minutes (20 minutes for the presentation + 10 minutes for questions and discussions). The venue of the presentation will be the

MIE Department Conference Room (2069) unless otherwise announced by the FYP coordinator. The presentation session will be open to any interested attendants.

The marking scheme is described as below:

No.	Activity	Weight %
1	Posture and confidence during the presentation	10
2	Clarity of presentation (Problem, objectives, constraints, Solution approach, results etc.)	10
3	Proper method of design analysis: Theoretical/Experimental (FYP2 only)	20
4	Rationality of achieved results, interpretation and conclusions (FYP2 only)	20
5	Ability to use visual aids and/or models effectively	10
6	Ability to handle questions	20
7	Ability to follow the time schedule of the presentation (FYP2 only)	10

Project Topics Profile

According to the policy of the department; conducting final year projects related to real industrial applications is highly encouraged. Although the projects are normally offered by faculty members, the Department encourages students to come up with their own topics of interest in consultation with a faculty member or an industrial partner in Oman. These could arise either from their industrial training experience or personal communication with an expert at an industrial firm. In this case, the student(s) should secure the approval of a faculty member who may be interested in the topic and is willing to supervise the project. As per the Accreditation Board for Engineering and Technology (ABET); <http://www.abet.org/>, requirements all offered projects have to include design element(s) in addition to economical, environmental and ethical aspects, in addition to the encouragement of projects of multi-disciplinary nature.

Project Coordinator

A staff member is nominated by the department to serve as FYP coordinator. The tasks of the FYP Coordinator include, but not limited to, the following:

1. Assure the implementation of the department policies for FYPs.
2. Develop a program schedule for project execution throughout the academic year.
3. Collect and organize the suggested project titles and submit them to the Department Board for approval.
4. Assign and announce the approved project titles to the students.
5. Keep the project application forms, evaluation forms and any related documents.
6. Organize the presentation sessions.
7. Collect the evaluation forms from supervisors and examiners, and work out the final grades.
8. Maintain a database archives which may include hard or/and soft copies of the FYP reports, logbooks, posters and presentations.

FYP Seminar Series

FYPs is a major element in the IE program ABET accreditation process and we, as a department, count on those FYPs to represent the quality of our graduates and in turn the program itself. Consequently; the Department is arranging a series of seminars to support all FYP students all through the project entire lifetime.

#	Seminar Subject
1	Final Year Project: An Overview (at the end of previous semester)
2	Report Writing, Referencing, Academic Integrity and Plagiarism.
3	ABET Accreditation: An Overview: Design Standards and Realistic Constraints
4	Handling Poster and Oral FYP Presentations

Project Assignment Summary (Template will be sent to Faculty members)

Project offered by faculty members should be accompanied by a short summary according to the format listed below. The short summary is needed for the following reasons:

- To help both the student(s) and the advisor in defining the project scope so that an appropriate open-ended problem is assigned.
- To clarify the objectives so that the student will be made accountable for achieving or working towards those objectives.
- To give an initial guideline so that the student will assess his/her capability against the given project requirements. This will help the students in making the right choice.
- To justify the number of students assigned to each project.

For projects suggested by students, a similar summary is prepared in collaboration with the FYP coordinator and/or the faculty member selected by the students to supervise the project.

Project Assignment Procedures

This procedure depends on the way the project has been initiated:

- **Projects offered by faculty members:** students need to fill the *Project Selection Form* (made available by the Coordinator) by selecting a maximum of five projects in the order of the group preference. The project assignment is made based on the average GPA of the group members; groups with higher average GPAs will be given the priority to select from the offered projects.
- **Project initiated by students:** the same form has to be filled by listing the project title and the name(s) of the supervisor(s). This must be done for referencing and the record only; students will automatically be assigned the project they suggested under the supervision of the selected faculty member.

Program Approval

Before the project titles and their short summaries are announced to the students, they must be approved by the Program Faculty members in consultation with the HoD. Projects that were approved in previous semesters but were not selected do not need to be approved if they will be reoffered.

FYP Students' Responsibilities

1. **FYP briefing and workshops:** Students should attend FYP briefing announced by the FYP coordinator to introduce FYP and any planned FYP seminars and workshops.
2. **Eligibility to Apply for FYP:** Before a student can apply for taking an FYP1n an academic year, he needs to ensure his eligibility to applying. Parameters determining the student's eligibility include: *total cumulative credit hours and prerequisites as specified in the degree plan.*
3. **Forming FYP Groups:** The students should form a group of 3-4 students. Each group of students should nominate a *group leader* who will be approached by the FYP coordinator for any arising matters during their work. The group leader, on behalf of his group, will be in charge of contacting the FYP coordinator, the supervisor and examining committee.
4. **FYP Application:** Each group should select a maximum of five choices from the FYPs offered in the Department and announced by the FYP coordinator. They need to submit their application, using the *Project Selection Form*, to the FYP coordinator by the announced deadline;
5. **Supervisor Meetings and Logbook:** Each group of students should arrange to meet their supervisor on a weekly or bi-weekly basis unless otherwise agreed with the project supervisor. It is the student responsibility to attend those meetings; *failure to show up will result in applying penalties to the student's final FYP grade.* Each group leader will be responsible for keeping details of each meeting and minutes (Logbook), organization of the work among group members, raising and discussing any faced problems with the supervisor and FYP coordinator. The leader will then complete the logbook form, collect the signature of the supervisor and submit it to the coordinator.
6. **Submissions of FYP proposal and Reports:** It is the student's responsibility to submit the required proposals, progress reports, draft and final report of the FYP *by their associated deadlines* mentioned in the FYP handbook. *Failure to comply with those deadlines may cause implementing penalties for each encountered delayed submission.*
7. **Tasks and Deliverables Accomplishment:** Each group *must fulfill and submit* all the tasks and deliverables required for the project (proposal, progress reports, logbook, draft and final report, etc.) using the right format and/or templates provided by the FYP coordinator by the deadlines mentioned in the FYP diary.
8. **Reports/Design Revision:** The students *must perform any changes*; amendments and/or modifications advised by the supervisor, on project deliverables (report, developed design/software), specified by the supervisor and/or the examining committee and resubmit their work by the advised deadline. *Failure to do so may result in an In Complete (IP) case of the FYP1n addition to facing penalties on the FYP final grade.*

FYP Supervisor's Check List

1. **Offering projects:** Offer final year project topics and verify that the topics offered are reasonable for degree-level final year projects. As per ABET requirements, the projects have

to include design element(s) in addition to economic, environmental and ethical dimensions. All offered FYPs should be submitted to the FYP coordinator by the deadline; week 11. Project assignment summary must be submitted to FYP coordinator by the faculty member offering a project before the Program Faculty members approval.

2. **Number of Supervised Projects:** Offer a sufficient number of projects every regular semester (as needed) and ensure that sufficient time will be allocated for each group throughout the semester. The total number of supervised projects will include both projects initiated/suggested by a group of students and FYPs proposed by the supervisor through normal ways of distribution of students based on their average GPAs.
3. **Students-Initiated Projects:** In case where students offer their own topics, the supervisor needs to verify that the topic proposed is reasonable for a degree-level final year project. The project assignment summary should be provided by both the supervisor and the students and then needs to be approved by the Program Faculty members. However, it will be assigned directly to the group initiated the topic.
4. **Meeting with Groups:** Set a weekly or bi-weekly meeting for each project and be available during the allocated time slot.
5. **Logbook Follow Up and Signature:** Sign the group's logbook for weekly/bi-weekly meeting. This would become the proof of the group attendance. If any student misses three meetings, without valid excuse, during the semester, the supervisor has the full authority, in consultation with the HoD, to bar the student who will get "F" at the end of the semester.
6. **Reports Correction:** Check and correct draft reports submitted by students before the final submission to the Coordinator and examining committee.
7. **Project Quality and Standards:** Verify that a group project has achieved the required standard before allowing them to present their final work. It is necessary to ensure that their FYP has clear realistic constraints, reflection on the real world, economic, environmental and ethical dimensions. The supervisor needs to verify that there is no plagiarism committed either in the FYP report, the application developed and/or the result obtained.
8. **Grading FYP:** Give marks to each deliverable and other criteria required for evaluation.
9. **Grades Submissions:** Submit project marks by the advised deadline to the FYP coordinator.

Logbook

Keeping a day to day record is a very useful practice when carrying out a major investigation task. Each project team should maintain a LOGBOOK to document all project activities; minutes of meetings with the supervisor or co-supervisors, library search bibliography, sketches of experimental setup designs, experimental data, any notes regarding the assignments given by the supervisor(s), ...etc. Each team should arrange for a weekly meeting with their supervisor(s) or as agreed with the project supervisor(s). Notes about the meeting should be recorded in the logbook in addition to the date of each entry. The students' supervisors are expected to see the logbook when meeting with students and this can show the supervisor how the project is progressing and what kind of guidance is needed. At the end of every meeting the leader must complete the logbook form (made available by the coordinator), collect the supervisors' signature and submit it to the coordinator within three working days.

Project Proposal

FYP1 students are required to submit a project proposal. A copy of this proposal is to be submitted first to the faculty supervisor for approval and should not exceed 5 pages.

Poster / Presentation

Students of FYP1 have to prepare a full scale poster, this is to be used in a special poster session in which all FYP students have to present their work. The poster size should be 100*70 cm and has no specific template. FYP2 students have to submit an electronic copy using Power Point (or similar) software package.

Examination Committee

The FYP coordinator will assign an examination committee for each project; each examining committee consists of two faculty members. It is recommended that the committee members to have expertise close to the project subject as per the availability at the time of the examination. After the supervisor approval of the project report, the examiners will receive their copies for evaluation. The results will be forwarded to the FYP Coordinator for processing the projects final grades. Time and dates of the examination is arranged by the FYP coordinator and announced to the whole Department. Both the examiners should attend the poster/presentation session.

Disclaimer Statement Format

The report is a document written by the student and should reflect his expertise in the different aspects of research methodology and technical writing skills. The supervisor's job is to guide the student so that he can achieve the objectives in an efficient way while gaining the skills sought. While maintaining credit, the *disclaimer statement* is simply a statement protecting the Department and the University from any legal liability claims associated with the use of the results and the methods presented. Its format is reported in the Report template.

Best Project Award

To encourage competition between FYP teams and creativity, a best project award is established. Few final year projects will be awarded a financial award of an amount determined based on the availability of funds which may come from the Department or from external sponsors. The selection procedure is as follow:

- Faculty members are asked to nominate the candidate projects.
- The Head of the Department forms a committee of three faculty members and two examiners from industry to screen and select the winning projects.

Program - FYP Rules and Policies

1. Deliverable Delayed Submissions: (Maximum penalty points of 10%)

It is the student's responsibility to submit the required proposal, progress report and final report of the FYP project by their associated deadlines mentioned in the FYP handbook. Failure to comply with the policies may cause implementing penalties for each encountered delayed submission. Maximum penalty points of 10% may be deducted from the final FYP grade per each student in case he/she encounters those delays. The FYP coordinator will be in charge to apply those penalties.

The detailed penalties scheme associated to any delayed actions is described in the following table:

Delayed action	Submitted to (Hard copy)	Due date	Penalty points	Remarks
Project proposal (FYP1)	FYP coordinator	Week 5	5%	Softcopy to FYP coordinator
Written Progress Report	FYP coordinator	Week 10	5%	Softcopy to FYP coordinator
Draft project report	Supervisor	<u>End of</u> Week 13	n/a	Softcopy to FYP coordinator
Corrected project report after implementing supervisor's comments/amendments	Both the examiners	<u>End of</u> Week 14	10%	Softcopy to FYP supervisor, examiners and coordinator
Draft poster design (FYP1)	Supervisor	<u>Start of</u> Week 14	5%	Softcopy to FYP coordinator
Poster presentation (FYP1)	FYP coordinator and supervisor	Week 15	5%	Softcopy to FYP coordinator
Oral presentation (FYP2)	FYP coordinator and supervisor	Week 15	10%	Softcopy to FYP coordinator

The deadlines for handing in draft report and final report must be followed strictly. In the case where a group needs an extension for report submission, the student needs to get approval from the FYP coordinator, the supervisor and the examining committee. If the student hands in the report later than the given deadline without asking for permission, the coordinator will apply the penalty policy for the associated delay.

2. Regular Meeting with Project Supervisor: (Maximum penalty points of 7.5%)

Each group of students should arrange to meet their supervisor on a weekly or bi-weekly basis unless otherwise agreed with the project supervisor. It is the student responsibility to attend those meetings; failure to attend on a regular basis or as advised by your supervisor may result in applying penalty points to a maximum of 7.5% (2.5% for each meeting) from the student FYP final grade. After two absentees the student will receive a warning from the Coordinator and after a total of FOUR absentees (without a valid excuse), then the coordinator, in consultation with the HoD, will bar the student from the group which means that his grade will be "F" at the end of the semester.

3. Submissions of the weekly Logbook Forms: (Maximum penalty points of 5%)

It is the responsibility of the project's leader to complete after every meeting the logbook form, to collect the signature of the supervisor and to submit a softcopy of it to the coordinator within three working days. Every two delays of logbook form submission will result in 2.5% penalty to the final grade for all the team members.

4. Final Submissions Allowances and Penalties:

Under very restricted circumstances, approved and confirmed by the project supervisor, the students may be given an extension period appropriate to the outstanding work. The extension period should be suggested by the project supervisor to the FYP coordinator and then approved by the HoD.

Under all circumstances, the maximum possible extension for submission of the project final report/poster is two weeks. This two-week extension will be accompanied by dropping the project final grade/student to the next lower grade; (A to A-), (A- to B+) etc. Any further extension, not to exceed one week at a time, will result in further dropping the new grade to the next lower grade and so on. In addition to the possible dropping of FYP final grade as mentioned above, students are advised that penalties tabulated in item 1 (Deliverable Delayed Submissions) are still applicable for each delayed submissions of any of the project deliverables.

The students are advised that the supervisor has the full authority NOT to grant any extension to the students if he thinks that they do not deserve any extension or in case there is no proper reason for the delay.

5. Failure to Attend the Presentation Session:

If a group fails to attend the presentation session allocated for them, the grade assigned to their presentation may be deducted totally unless they have a very valid reason for not showing up. This should be through a formal documented excuse from a governmental institution and approved by the Assistant Dean for Undergraduate students. This letter should be submitted to the FYP coordinator and the project supervisor within a maximum of three days of the scheduled presentation date. The group may be scheduled for another presentation by the FYP coordinator in consultation with the examining committee.

6. Change of Project:

Any group may be allowed to change their project by the end of Week 2. This should be with a written agreement from the current supervisor mentioning reason(s) of such a change. This change only applies to FYP1; changing FYP2 project is NOT allowed under any circumstances. Students will not be allowed to change supervisors unless they get the approval of the FYP coordinator who needs to seek a confirmation from the HoD.

7. Act of Dishonesty:

If a group is caught cheating in their FYP project, the FYP coordinator, supervisor and the examining committee have the full authority to fail this project after the approval of the HoD within a maximum of three working days from the date of examination. The following are representing possible cheating behaviors:

- Asking someone else to do the assigned work.
- Copying other people's report/thesis without proper citation in the reference section.
- Copying other people's report/thesis from websites.
- Plagiarism in project report/thesis.
- Faking the survey or questionnaires results.

8. Supervisor Unavailability:

If the project supervisor suddenly becomes unavailable (research leave, sabbatical leave, warded or passing away), the FYP coordinator should arrange, within One week, with the HoD to nominate the 1st co-supervisor (if applicable) or nominate a new supervisor in the same area where possible.